There are two questions we might ask about our harmonic oscillator:

- How does the state-vector change as a function of time? To answer this question, we need to know the Hamiltonian.

- What are the oscillator's possible energies? These are also determined by the Hamiltonian.

So to know anything useful we need the Hamiltonian. Fortunately, we can derive it from the Lagrangian, and I'll remind you how in a moment. But first recall that the canonical momentum conjugate to x is defined as $\partial L/\partial \dot{x}$[1] Combining this with Eq. 10.5, we get

$$p = \frac{\partial L}{\partial \dot{x}} = \dot{x}.$$

Using the straightforward definition from classical mechanics, we find that the Hamiltonian for the harmonic oscillator is

$$H = p\dot{x} - \mathcal{L},$$

where p is the canonical momentum conjugate to x, and $\mathcal{L}$ represents the Lagrangian.[2] We could work directly from this definition, but instead we'll take a shortcut. Because the Lagrangian is the kinetic energy *minus* the potential energy, the Hamiltonian is the kinetic energy *plus* the potential energy—in other words, the total energy. The Hamiltonian for the oscillator can therefore be written

$$H = \frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2.$$

So far, so good, but we're not quite finished. We've expressed kinetic energy in terms of velocity; in quantum mechanics, however, we need to represent our observables as *operators*, and we don't have a velocity operator. To take